Neel Gali

Engineer, Model Deployment & Inference Solutions

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Epik Solutions | Real-time AI/ML Inference Platform

- Designed **Python** and **C++** services for a high-performance **ML deployment platform** on **AWS** to manage model rollouts across 500k+ autonomous vehicles, achieving 99.9% uptime.
- Implemented **model-optimization experiments** using **PyTorch** for **quantization** and **pruning** on **on-vehicle hardware**, reducing model sizes by 40% while preserving accuracy within 1%.
- Developed **agentic specialists** with **LangGraph** and **OpenAI API** for automating **production deployment** health checks, decreasing manual intervention by 75% for critical AV stack updates.
- Built **platform tools** including **performance probes** and **validators** in **Go** and **Python** to ensure **real-time latency** and **memory budgets** compliance for on-vehicle **ML inference**.
- Engineered **CI/CD pipelines** using **Jenkins** and **Docker** for automated testing and deployment of **Super Cruise** software, accelerating releases by 30% while maintaining safety standards.
- Collaborated with **AV organization** teams (kernels, compiler) to diagnose and resolve **production deployment** and **performance issues** using **Linux profiling tools**, improving system stability by 20%.
- Spearheaded design and implementation of **secure coding** and **compliance practices** for **eyes-off autonomous driving** systems, successfully passing 100% of internal and external security audits.

Software Engineer *Jul 2022 – Jun 2025*

Samay Systems LLC | High-Performance ML Operations Platform

- Developed **Python**-based **ML systems** for **deep learning** model training and serving, processing 5TB of data daily, improving model refresh rates by 50%.
- Engineered scalable **data pipelines** using **Apache Kafka** and **Spark** on **AWS EMR** to ingest 10M events/second, supporting real-time analytics for 100k active users.
- Built **RESTful APIs** with **FastAPI** and **PostgreSQL** for **ML model serving**, handling 10,000 requests per second with average latency under 50ms.
- Integrated **OpenAI API** for natural language processing tasks into customer-facing applications, enhancing user engagement metrics by 25% through improved content generation.
- Optimized **TensorFlow** and **Keras computer vision** models for inference speed on **GPU** clusters, achieving 2x faster processing for image classification tasks.
- Managed **Docker** containers and **Kubernetes** clusters for **ML microservices** deployments, reducing infrastructure costs by 15% through efficient resource utilization.
- Implemented monitoring and alerting systems using **Prometheus** and **Grafana** for **production deployment** health, proactively resolving 90% of issues before user impact.
- Designed and implemented **secure coding practices** and **access control mechanisms** for **sensitive data** within the **ML platform**, achieving 100% compliance with industry standards.

Software Engineer *Jul 2020 – Aug 2022*

Zemoso Technologies | Custom Software Solutions & API Development

- Developed robust **Java Spring Boot** microservices for enterprise clients, handling 5,000 concurrent users with an average response time of 100ms.
- Built dynamic user interfaces using **React.js** and **JavaScript**, improving client application responsiveness by 30% and user satisfaction scores by 15%.
- Designed and managed **PostgreSQL** and **MongoDB** databases for various projects, ensuring data integrity and optimizing queries for 2x faster retrieval.
- Implemented **RESTful APIs** for third-party integrations, increasing platform extensibility and enabling new partnerships that generated $500k in annual revenue.
- Contributed to **CI/CD pipelines** using **Jenkins** and **Git** for automated code deployments, reducing manual release efforts by 40% across various projects.

Skills

Languages: Python, C++, Go, Java, JavaScript
Frameworks & Tools: PyTorch, TensorFlow, Keras, FastAPI, Spring Boot, React.js, Docker, Kubernetes, Apache Kafka, Spark, Jenkins, Git, Prometheus, Grafana, LangGraph, OpenAI API
Cloud & Infra: AWS, AWS EMR, Linux, PostgreSQL, MongoDB
Concepts: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, ML Systems, Computer Architecture, Operating Systems, Distributed Systems, Model Optimization, Quantization, Pruning, Distillation, Real-time Systems, Scalability, API Design, System Design, Secure Coding, Safety, Compliance, Microservices, Data Structures, CI/CD, Automated Testing, Observability, Backend Development, Frontend Development, UI/UX

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India